

Pseudo-Laplacians at Whitney Cross-Caps: Point Interactions on Surfaces Mapped into Three-Space

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Abstract

On a complete smooth Riemannian surface, Colin de Verdière showed that the Laplacian restricted to functions vanishing near a point admits a one-parameter family of self-adjoint extensions, the pseudo-Laplacians, whose domains are described by a logarithm and a constant. We show that the same picture holds at the Whitney cross-caps of a surface mapped into $\mathbb{R}^3$. These are the generic singularities of such maps, and the induced metric degenerates at them. Each cross-cap carries exactly one logarithmic channel. The leading coefficient is the universal $-1/(2\pi)$ in the intrinsic distance, or equivalently in the extrinsic radius, and the regular part is Hölder continuous. With k cross-caps, the extensions of the point-restricted Laplacian are classified by $U(k)$ in explicit geometric boundary coordinates with a 2π Green pairing. The Friedrichs extension is the unique extension with a sub-Markovian semigroup, and the unique one invariant under change of logarithmic reference length. The key step is an arclength coordinate that reveals an asymptotically Euclidean geometry at the cross-cap: the metric is the identity up to $O(\rho^{1/2})$, which refines Grieser's quasi-isometry of the Whitney umbrella. Examples include Steiner's Roman surface: its six cross-caps carry a $U(6)$ -family of pseudo-Laplacians, and octahedral symmetry forces its Green matrix to have at most three eigenvalues.

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1 Introduction

Let X be a complete smooth Riemannian surface and $x_0 \in X$. Colin de Verdière [1] observed that the Laplacian on $C_c^\infty(X \setminus \{x_0\})$ is not essentially self-adjoint. Its deficiency indices are $(1, 1)$, and the self-adjoint extensions, the *pseudo-Laplacians*, are singled out by the behaviour

$$f(x) = \lambda \left(\sin \alpha \frac{1}{2\pi} \log r + \cos \alpha \right) + o(1), \quad r = d(x, x_0).$$

This picture underlies point interactions on manifolds. For conical points of a flat surface [2], the logarithm is again the only channel when the cone angle is at most 2π ; larger angles add power-law channels.

The singularities of a generic smooth map of a surface into $\mathbb{R}^3$, when present, are isolated *cross-caps* [4, 5], and they are stable under small perturbations. At a cross-cap the differential has rank one, and the induced metric $g = f^*\delta$ degenerates. This paper asks what happens to point interactions at these points. The answer is pleasantly simple: for the scalar Laplacian, a cross-cap behaves like a regular point of the surface. Although the induced metric degenerates there, an arclength coordinate reveals an asymptotically Euclidean geometry (Lemma 3.2, Figure 1). This allows the classical logarithmic point-interaction picture to persist at the singularity, with its exact coefficient.

Main results. Let $f : \Sigma \rightarrow \mathbb{R}^3$ be a smooth map of a compact surface, immersive except at finitely many cross-caps; the precise setting is Definition 2.1. Let H_F be the Friedrichs Laplacian of g , with Dirichlet condition if Σ has a boundary, and let $p_1, \dots, p_k$ be the interior cross-caps.

1. H_F has compact resolvent (Theorem 4.2). Every $u \in \text{Dom } H_F$ is continuous at each p_j , and $\tau u = (u(p_j))_j$ is a graph-bounded surjection onto $\mathbb{C}^k$ with dense kernel (Theorem 5.1).
2. The restriction $A = H_F|_{\ker \tau}$ has deficiency indices (k, k) , and its self-adjoint extensions form a $U(k)$ -family with explicit Krein resolvent formula (Theorem 5.2).
3. Every $\varphi \in \text{Dom } A^*$ has the expansion $\varphi = \ell_j \log d_g(\cdot, p_j) + b_j + O(d_g^\beta)$ at each p_j , and the Green form is $2\pi(\langle \ell, b' \rangle - \langle b, \ell' \rangle)$. In particular the resolvent Green vectors have leading term $-\frac{1}{2\pi} \log d_g$ (Theorems 6.1 and 6.3).
4. H_F is the only extension with a sub-Markovian semigroup, and the only one invariant under a common change of logarithmic reference length. Coefficients on the two sheets meeting at a cross-cap agree in the limit (Section 7).

Each cross-cap therefore carries one channel, as a regular point does, and its Whitney invariants enter only through the regular part (Table 1). The geometric input is the arclength coordinate of Lemma 3.2, in which the metric equals the identity up to $O(\rho^{1/2})$. This refines Grieser's quasi-isometry of the Whitney umbrella with the plane [3] from bounded distortion to asymptotic equality, and it is this step that fixes the exact coefficient $-1/(2\pi)$.

The theory is local at the cross-caps but needs a global boundary condition. If one instead closes the Laplacian on functions compactly supported in the interior, every regular boundary arc contributes infinitely many deficiency modes. The Krein extension of that closure then has an infinite-dimensional kernel, while its nonzero spectrum is still discrete and is given by a buckling problem (Section 8). The examples in Section 9 are Steiner's Roman surface, a closed surface with six cross-caps, and a ruled surface from an earlier signal model, which also has cross-caps on its boundary.

Point	Total angle	Singular channels at the point	Source
Regular point of a complete smooth surface	2π	$\log r$	[1]
Conical point of a flat surface	$\theta \leq 2\pi$	$\log r$	[2]
Conical point of a flat surface	$\theta > 2\pi$	$\log r$ and $r^{- \nu }e^{i\nu\phi}$, $\nu = 2\pi j/\theta$, $0 < j < \theta/2\pi$	[2]
Whitney cross-cap	2π	$\log r$	this paper

Table 1: Singular channels of the point-restricted Laplacian. At a cross-cap the total angle is 2π (Remark 6.2), and a single logarithmic channel appears, exactly as at a regular point.

Relation to the literature. The abstract extension theory is Posilicano’s [7], and the scalar regularity input is the De Giorgi–Nash–Moser theory in the form of [9]. Colin de Verdière [1] treats regular points of smooth surfaces and Hillairet–Kokotov [2] conical points. Grieser [3] supplies the quasi-isometric coordinates. The contribution here is to carry this circle of ideas to the generic singular points of surfaces in $\mathbb{R}^3$, together with the sharpened local geometry that this requires.

2 Setting

Definition 2.1. Let Σ be a compact connected smooth surface, either closed or with a piecewise smooth Lipschitz boundary, and let $f : \Sigma \rightarrow \mathbb{R}^3$ be smooth up to the boundary. Assume that f is an immersion except on a finite set $Z \cup Z_\partial$, where $Z \subset \text{int } \Sigma$ and $Z_\partial \subset \partial\Sigma$, and that every point of $Z \cup Z_\partial$ is a *Whitney cross-cap* of f , or of a smooth extension of f near $\partial\Sigma$. That is, df has rank one there, and for a kernel vector η and a complementary vector ξ ,

$$\det(f_\xi, f_{\xi\eta}, f_{\eta\eta}) \neq 0. \quad (1)$$

At each $p \in Z_\partial$ we assume that, in some smooth chart of the extension centred at p , Σ is a closed sector $\{\theta_1 \leq \theta \leq \theta_2\}$. We write $Z = \{p_1, \dots, p_k\}$. There are two regimes:

- (D) $\partial\Sigma \neq \emptyset$, with the Dirichlet condition imposed on the whole boundary;
- (N) Σ is closed, in which case $Z_\partial = \emptyset$.

The condition (1) is the standard recognition criterion for the cross-cap $(x, y) \mapsto (x, xy, y^2)$ [6]. A theorem of Whitney [4], see also [5], states that the cross-cap is the only stable singularity of maps from surfaces to $\mathbb{R}^3$. Consequently, for a generic map the hypotheses hold with $Z_\partial = \emptyset$.

Put $g = f^*\delta$ on $\Sigma \setminus (Z \cup Z_\partial)$. Let $\mathcal{H} = L^2(\Sigma, dA_g)$ and $q(u, v) = \int \langle \nabla u, \nabla v \rangle_g dA_g$. Take as core $\mathcal{C} = C_c^\infty(\text{int } \Sigma \setminus Z)$ in case (D), and the smooth functions on Σ vanishing near Z in case (N). The closure $\mathcal{V}$ of $\mathcal{C}$ in the norm $(q(u, u) + \|u\|^2)^{1/2}$ defines a closed form, and we let H_F be its self-adjoint operator. Scalar products are conjugate-linear in the first slot.

All operators act on the source. Functions live on Σ , and f enters only through the pulled-back metric g , its area form and its gradient. Where $f(s) = f(s')$ for distinct source points, as along the double curves of the Roman surface, the two points remain distinct, and a function may take different values on the two sheets. Thus “the Laplacian of the Roman surface” means the Laplacian of $(\mathbb{RP}^2, f^*\delta)$. The self-intersections of the image play no role; the cross-caps, where g degenerates, are what matter.

3 Local geometry at a cross-cap

Lemma 3.1 (Adapted form). *After a translation and a rotation of $\mathbb{R}^3$ and a smooth change of source coordinates centred at a cross-cap, $f(x, y) = (x, h(x, y))$ with $h(0) = 0$, $Dh(0) = 0$ and*

$$h_y(x, y) = xv + 2yw + O(R^2), \quad R = (x^2 + y^2)^{1/2},$$

where $v, w \in \mathbb{R}^2$ are linearly independent. Hence $cR \leq |h_y| \leq CR$ and $|h_x| \leq CR$. In any smooth chart centred at the point, $g \leq CI$ and $cR^2 \leq \det g \leq CR^2$; in particular $dA_g \asymp R dx dy$.

Proof. Rotate so that $f_\xi(0)$ points along the first axis. Use x , the first component of f , together with any coordinate y completing it to a chart. Since the image of $df(0)$ is the first axis, $\ker dx(0) = \ker df(0)$, so ∂_y is a kernel vector at 0. Then $h_x(0) = 0$ and $h_y(0) = 0$. The vectors $v = h_{xy}(0)$ and $2w = h_{yy}(0)$ are independent by (1). In these coordinates $g = \begin{pmatrix} 1+|h_x|^2 & h_x \cdot h_y \\ h_x \cdot h_y & |h_y|^2 \end{pmatrix}$, whose entries are bounded and whose determinant lies between $|h_y|^2$ and $(1 + |h_x|^2)|h_y|^2$. A smooth change of chart has bounded derivative and inverse, so these bounds persist. $\square$

Lemma 3.2 (Arclength coordinate). *In the coordinates of Lemma 3.1 put*

$$V(x, y) = \int_0^y |h_y(x, s)| ds, \quad \Xi(x, y) = (x, V), \quad \rho = |\Xi|.$$

Then Ξ is a homeomorphism of a neighbourhood of 0 onto a neighbourhood of 0, and a C^1 diffeomorphism off 0. In the coordinates $(X, V) = \Xi$, the metric G , the density $m = \sqrt{\det G}$ and the divergence tensor $a = mG^{-1}$ satisfy

$$G - I = O(\rho^{1/2}), \quad a - I = O(\rho^{1/2}), \quad m - 1 = O(\rho),$$

and are bounded and uniformly elliptic. The extrinsic radius $\rho_e = |f - f(0)|$ satisfies $\rho_e/\rho = 1 + O(\rho^{1/2})$.

Proof. For fixed x , V is strictly increasing in y , and $cy^2 \leq |V| \leq C(|xy| + y^2)$. Its inverse is therefore continuous, also at zero. Differentiating under the integral, with $n = h_y/|h_y|$, gives $V_y = |h_y|$ and $V_x = \int_0^y n \cdot h_{xy} ds$, so $|V_x| \leq C|y|$. Both are continuous, and $V_y > 0$ off zero. In the new coordinates $f_X = (1, q)$ and $f_V = (0, n)$, where $q = h_x - V_x n$, so

$$G = \begin{pmatrix} 1 + |q|^2 & q \cdot n \\ q \cdot n & 1 \end{pmatrix}, \quad \det G = 1 + |q|^2 - (q \cdot n)^2 \geq 1.$$

Since $|q| \leq CR$ and $R^2 \leq C\rho$, the stated bounds follow.

For the radii, Taylor's formula gives $|h - wy^2| \leq C(x^2 + |xy| + |y|R^2)$. Integrating $||h_y(x, s)| - 2|w||s|| \leq C(|x| + x^2 + s^2)$ gives $|V - \operatorname{sgn}(y)|w|y^2| \leq C(|xy| + |y|R^2)$. Hence $||h| - |V|| \leq CR\rho$, and comparing the norms of $(x, |h|)$ and $(x, |V|)$ gives $|\rho_e - \rho| \leq CR\rho \leq C\rho^{3/2}$. $\square$

Remark 3.3 (The standard cross-cap). For $f(x, y) = (x, xy, y^2)$ one has $h_y = (x, 2y)$, and the arclength coordinate is explicit:

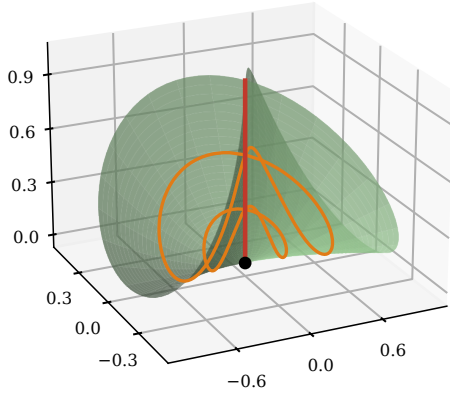
$$V(x, y) = \frac{y}{2} \sqrt{x^2 + 4y^2} + \frac{x^2}{4} \operatorname{arsinh} \frac{2y}{|x|}, \quad V(0, y) = y|y|.$$

Along the double ray $x = 0$, V is the signed arclength of the parabola $y \mapsto (0, 0, y^2)$, so the two sheets meeting there are unfolded in opposite directions (Figure 1). Here the general bound improves. With q and n as in the proof of Lemma 3.2,

$$V_x = \frac{x}{2} \operatorname{arsinh} \frac{2y}{|x|}, \quad q \cdot n = \frac{x}{2} \left(\frac{2y}{\sqrt{x^2 + 4y^2}} - \operatorname{arsinh} \frac{2y}{|x|} \right), \quad |q| \leq 2|y|.$$

Since $|x| \leq \rho$, $y^2 \leq |V| \leq \rho$, and $x \mapsto x \operatorname{arsinh}(a/x)$ is increasing on $(0, \infty)$ for $a > 0$, we get $|q \cdot n| \leq \frac{\rho}{2}(1 + \operatorname{arsinh}(2\rho^{-1/2}))$ and $|q|^2 \leq 4\rho$. Hence $G - I = O(\rho \log(1/\rho))$ for the standard germ. Figure 2 shows this rate numerically, for the metric and for the radius ratio.

(a) $f = (x, xy, y^2)$, double ray, links $\rho_e = r$



(b) levels of V (teal), $X = x$ (grey), ρ (dashed)

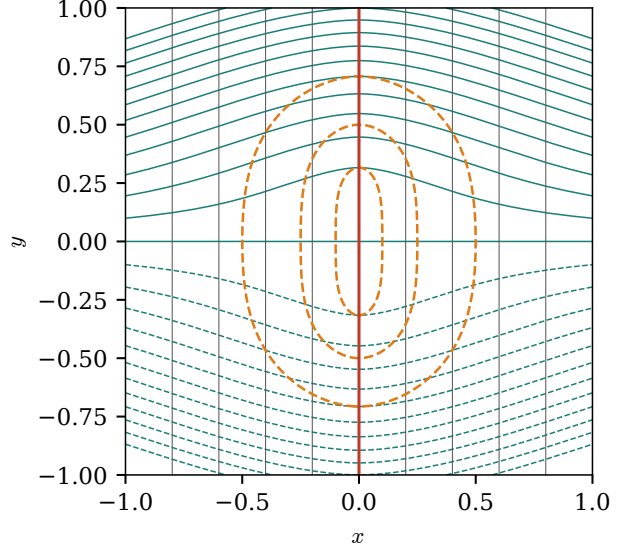


Figure 1: (a) The standard cross-cap $f(x, y) = (x, xy, y^2)$ on the unit disc of the source. The two sheets cross along the double ray (red), which ends at the cross-cap (black). The orange curves are links $\rho_e = r$, each the image of a circle with one double point. (b) The source in the arclength coordinates of Remark 3.3: level curves of V (teal, dashed where $V < 0$), of $X = x$ (grey), and the circles $\rho = 0.1, 0.25, 0.5$ (orange). In these coordinates the degenerate metric becomes asymptotically Euclidean.

Corollary 3.4 (Intrinsic distance). *Let d_g be the path distance of g on the source. Near a cross-cap p ,*

$$d_g(\cdot, p) = \rho(1 + O(\rho^{1/2})), \quad \log d_g(\cdot, p) - \log \rho_e = O(\rho^{1/2}).$$

Proof. In the arclength coordinates, $|v|^2(1 - C\rho^{1/2}) \leq v^T G v \leq |v|^2(1 + C\rho^{1/2})$. The straight segment to a point at radius ρ has length at most $\rho(1 + C\rho^{1/2})$. A curve from p to that point either stays in the disk of radius 2ρ , and then has length at least $\rho(1 - C\rho^{1/2})$, or crosses the annulus $\rho \leq |\cdot| \leq 2\rho$, which costs at least $\rho(1 - C\rho^{1/2})$. Lemma 3.2 compares ρ with ρ_e . $\square$

Remark 3.5. Grieser proved that the Whitney umbrella metric is quasi-isometric to the Euclidean plane through a singular coordinate change [3, §1]. Lemma 3.2 sharpens this from bounded distortion to $G = I + O(\rho^{1/2})$. Bounded distortion would suffice for continuity of point values, but the exact constant $-1/(2\pi)$ requires the sharper form.

4 The Friedrichs Laplacian

Proposition 4.1 (Zero capacity). *Each interior cross-cap has zero capacity for q . Consequently $\mathcal{V}$ is also the closure of $C_c^\infty(\text{int } \Sigma)$ in case (D), or of $C^\infty(\Sigma)$ in case (N).*

Proof. In the arclength coordinates, q and the L^2 norm on a small disk are equivalent to the Euclidean H^1 and L^2 norms. Cutoffs equal to $\log(\varepsilon/\rho)/\log(1/\varepsilon)$ for $\varepsilon^2 \leq \rho \leq \varepsilon$, suitably smoothed, have Euclidean Dirichlet energy $O(1/\log(1/\varepsilon))$. Multiplying a smooth function by one minus such a cutoff therefore approximates it in the form norm by functions vanishing near the point. Smooth source functions have finite energy, because $|\nabla u|_g^2 dA_g = O(R^{-1}) dx dy$ by Lemma 3.1. $\square$

Theorem 4.2 (Compactness). *The embedding $\mathcal{V} \hookrightarrow \mathcal{H}$ is compact, so H_F has compact resolvent. In case (D), $H_F \geq \lambda_D > 0$. In case (N), $\ker H_F$ consists of the constants.*

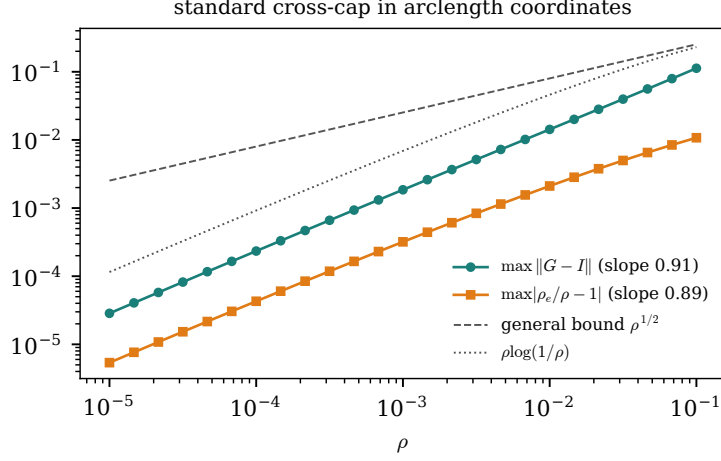


Figure 2: Deviation from Euclidean geometry at the standard cross-cap, sampled at 720 points of each circle $\rho = r$ in arclength coordinates. The metric deviation $\|G - I\|$ and the radius ratio ρ_e/ρ both approach the Euclidean values at the rate $\rho \log(1/\rho)$, well inside the general bound $\rho^{1/2}$ of Lemma 3.2.

Proof. Fix $p \in Z \cup Z_\partial$ and chart coordinates in which a neighbourhood of p in Σ is a sector: the full disk for interior points, and the sector of Definition 2.1 for boundary points. Let χ be a chart cutoff, equal to 1 near p , and let $u \in \mathcal{C}$, so that $h = \chi u$ vanishes near p . On each ray, integrating $(R|h|^2)'$ and applying Cauchy–Schwarz gives

$$\int_0^{R_0} |h|^2 dR \leq 4 \int_0^{R_0} R^2 |\partial_R h|^2 dR.$$

Integrating over the angular interval and using Lemma 3.1 ($dA_g \asymp R dx dy$ and $g \leq CI$),

$$\int \frac{|h|^2}{R^2} dA_g \leq C \int R |\nabla h|^2 dx dy \leq C' (q(u, u) + \|u\|^2).$$

This extends to $\mathcal{V}$ by closure. Hence the L^2 mass in $\{R < \varepsilon\}$ is at most $C\varepsilon^2$ times the form norm, uniformly over the finitely many singular points. On the complement of these small sectors, g is a smooth Riemannian metric on a compact Lipschitz domain, where Rellich’s theorem applies. A diagonal argument together with the uniform small-sector bound gives compactness.

If $q(u_n, u_n) \rightarrow 0$ with $\|u_n\| = 1$, a subsequence converges in $\mathcal{H}$ to a function with zero gradient on the connected regular locus, which is a constant. In case (D) its Dirichlet trace on a regular boundary arc vanishes, so the constant is zero, a contradiction. In case (N) the constants lie in $\mathcal{V}$ and form the kernel. $\square$

5 Point values and the extension problem

Theorem 5.1 (Point traces). *Each $u \in \text{Dom } H_F$ has a unique representative that is continuous at every p_j , Hölder continuous in the arclength coordinates. The map $\tau u = (u(p_1), \dots, u(p_k))$ satisfies $|\tau u| \leq C(\|u\| + \|H_F u\|)$, is onto $\mathbb{C}^k$, and has kernel dense in $\mathcal{H}$.*

Proof. Near p_j , in the arclength coordinates, u is a local H^1 weak solution of $-\text{div}(a \nabla u) = m H_F u$. This equation holds across the point because, by Proposition 4.1, test functions vanishing near the point are dense in H^1 . The coefficients are real, bounded and uniformly elliptic, and $m H_F u \in L^2$ with $2 > \dim/2$, so the interior Hölder estimate [8, Thm. 8.24], [9, Lect. 18] gives $\|u\|_{C^\alpha} \leq C(\|u\|_{L^2} + \|H_F u\|_{L^2})$ on a smaller disk. A cutoff χ_j equal to 1 near p_j , supported

away from the other singular points and from $\partial\Sigma$, has $\Delta_g\chi_j$ smooth and supported on a regular annulus, so $\chi_j \in \text{Dom } H_F$ and $\tau\chi_j = e_j$. Finally the core $\mathcal{C}$ consists of smooth functions vanishing near Z (and near $\partial\Sigma$ in case (D)); integration by parts on the regular locus puts it in $\text{Dom } H_F \cap \ker \tau$, and it is dense in $\mathcal{H}$. $\square$

Theorem 5.2 (The point-restricted Laplacian). *The operator $A = H_F|_{\ker \tau}$ is closed, densely defined and symmetric, with deficiency indices (k, k) , and its Friedrichs extension is H_F . Fix $\eta < 0$, let $R_z = (H_F - z)^{-1}$, and put $G_z = (\tau R_z)^* : \mathbb{C}^k \rightarrow \mathcal{H}$. Then*

$$\text{Dom } A^* = \text{Dom } H_F \dot{+} G_\eta \mathbb{C}^k, \quad A^*(u + G_\eta a) = H_F u + \eta G_\eta a.$$

The maps $\Gamma_0(u + G_\eta a) = a$ and $\Gamma_1(u + G_\eta a) = \tau u$ form a boundary triple. The self-adjoint extensions are exactly the operators $(U - I)\Gamma_1 + i(U + I)\Gamma_0 = 0$, $U \in U(k)$, and their resolvents differ from R_z by operators of rank at most k . All of them have compact resolvent. In case (D), the Krein-von Neumann extension has domain $\text{Dom } A \dot{+} G_0 \mathbb{C}^k$ and kernel of dimension k .

Proof. Graph continuity makes $\ker \tau$ graph-closed. For nonreal z , the bounded surjection $\tau R_z : \mathcal{H} \rightarrow \mathbb{C}^k$ has kernel $\text{ran}(A - z)$, so $\dim \ker(A^* - \bar{z}) = k$. The form core $\mathcal{C}$ lies in $\text{Dom } A$, so the Friedrichs extension of A is H_F . For $\varphi \in \text{Dom } A^*$, the vector $u = R_\eta(A^* - \eta)\varphi$ lies in $\text{Dom } H_F$, and $\varphi - u$ lies in $\ker(A^* - \eta) = \text{ran}(A - \eta)^\perp = G_\eta \mathbb{C}^k$; the sum is direct because $\eta \in \rho(H_F)$. The Green identity

$$\langle A^* \varphi, \psi \rangle - \langle \varphi, A^* \psi \rangle = \langle \Gamma_1 \varphi, \Gamma_0 \psi \rangle - \langle \Gamma_0 \varphi, \Gamma_1 \psi \rangle$$

follow from $\langle (H_F - \eta)u, G_\eta b \rangle = \langle \tau u, b \rangle$. This is the framework of [7]. Resolvent differences have range in $\ker(A^* - z)$, which has dimension k , and R_z is compact. In case (D) A is strictly positive, and the Krein theorem [10, Thm. 2.1] gives $\text{Dom } A_K = \text{Dom } A \dot{+} \ker A^*$, with $\ker A^* = G_0 \mathbb{C}^k$ since $0 \in \rho(H_F)$. $\square$

Corollary 5.3 (Eigenvalue counting). *Every self-adjoint extension H of A is bounded below and has compact resolvent. Its eigenvalue counting function satisfies $|N_H(\lambda) - N_{H_F}(\lambda)| \leq k$ for all real λ .*

Proof. Extensions of a semibounded symmetric operator with finite deficiency indices are semibounded. Choose μ below both spectra. Then $(H - \mu)^{-1}$ and $(H_F - \mu)^{-1}$ are positive compact operators whose difference has rank at most k . By Weyl's interlacing inequalities, their eigenvalue counting functions above any threshold differ by at most k . The map $\lambda \mapsto (\lambda - \mu)^{-1}$ is decreasing on (μ, ∞) , which transfers the bound to N_H and N_{H_F} . $\square$

6 Logarithmic normalization

Theorem 6.1 (Green vectors). *For real $\eta < 0$ there are $\beta > 0$ and a real symmetric matrix B_η such that, near each p_i ,*

$$G_{\eta,j} = -\frac{\delta_{ij}}{2\pi} \log d_g(\cdot, p_i) + (B_\eta)_{ij} + O(d_g(\cdot, p_i)^\beta).$$

The same holds with d_g replaced by the extrinsic radius $\rho_{e,i} = |f - f(p_i)|$, with the same matrix B_η .

Proof. Fix i and work in arclength coordinates X on a disc B_{2r_0} about p_i , with r_0 so small that these discs are disjoint for different poles. There $\rho = |X|$, $dA_g = m dX$, $q(u, v) = \int a \nabla \bar{u} \cdot \nabla v dX$, and $E = a - I$ satisfies $|E| \leq C\rho^{1/2}$ by Lemma 3.2. Since q and η are real, H_F commutes with complex conjugation, and it suffices to pair with real functions.

Step 1: a parametrix. Let χ be a radial cutoff equal to 1 on B_{r_0} and supported in B_{2r_0} , and put $h_0 = -\chi \log \rho / (2\pi)$. Then $-\Delta h_0 = \delta_0 + k_0$, with k_0 bounded and supported in the annulus $r_0 \leq \rho \leq 2r_0$. For $\varphi \in \mathcal{V}$ put

$$\mathcal{F}(\varphi) = \int E \nabla h_0 \cdot \nabla \varphi dX + \int (k_0 - \eta m h_0) \varphi dX.$$

Since $|E \nabla h_0| \leq C \rho^{-1/2}$, the first density lies in L^3 , and $k_0 - \eta m h_0$ lies in L^2 . By uniform ellipticity $\mathcal{F}$ is bounded on $\mathcal{V}$. The form $q_\eta = q - \eta \langle \cdot, \cdot \rangle$ is coercive, so Lax–Milgram gives a real $w \in \mathcal{V}$ with $q_\eta(w, \varphi) = \mathcal{F}(\varphi)$ for all $\varphi \in \mathcal{V}$. Locally, w solves a uniformly elliptic equation with divergence data in L^3 and scalar data in L^2 , so the Hölder estimate [8, Thm. 8.24], [9, Lect. 18] makes w Hölder continuous near every pole. Put $g_i = h_0 - w$.

Step 2: an annulus estimate. Let $u \in \text{Dom } H_F$ be real. By Theorem 5.1, $|u - u(p_i)| \leq C \rho^\alpha$ on B_{r_0} . Testing $-\text{div}(a \nabla u) = m H_F u$ with $\zeta^2(u - u(p_i))$, where ζ equals 1 on B_r , is supported in B_{2r} and has $|\nabla \zeta| \leq C/r$, gives the Caccioppoli bound

$$\int_{B_r} |\nabla u|^2 \leq \frac{C}{r^2} \int_{B_{2r}} |u - u(p_i)|^2 + C \int_{B_{2r}} |H_F u| |u - u(p_i)| \leq C r^{2\alpha}.$$

On an annulus $A_r = B_{2r} \setminus B_r$ we have $|\nabla h_0| \leq C/r$ and $|A_r| \leq C r^2$, so Cauchy–Schwarz gives $\int_{A_r} |\nabla h_0| |\nabla u| \leq C r^\alpha$. Summing over dyadic annuli shows that $\nabla h_0 \cdot \nabla u$ is integrable on B_{r_0} . Now let ζ_ε be radial, equal to 0 on B_ε and to 1 off $B_{2\varepsilon}$, with $|\nabla \zeta_\varepsilon| \leq C/\varepsilon$. The same bounds give

$$\int |h_0| |\nabla \zeta_\varepsilon| |\nabla u| \leq C \varepsilon^\alpha |\log \varepsilon|, \quad \int |u - u(p_i)| |\nabla h_0| |\nabla \zeta_\varepsilon| \leq C \varepsilon^\alpha.$$

Both tend to zero: excising a small disc costs nothing in the limit.

Step 3: identification. Put $v = (H_F - \eta)u$. Since $\zeta_\varepsilon h_0 \in \mathcal{V}$,

$$\langle \zeta_\varepsilon h_0, v \rangle = q_\eta(\zeta_\varepsilon h_0, u) = \int a \nabla(\zeta_\varepsilon h_0) \cdot \nabla u dX - \eta \int m \zeta_\varepsilon h_0 u dX.$$

As $\varepsilon \rightarrow 0$, Step 2 gives $\langle h_0, v \rangle = \int a \nabla h_0 \cdot \nabla u dX - \eta \int m h_0 u dX$. For the Euclidean part of $a = I + E$, integrate by parts on the support of ζ_ε , where h_0 is smooth, and use $\int \nabla h_0 \cdot \nabla \zeta_\varepsilon dX = -1$, a direct computation in polar coordinates:

$$\int \zeta_\varepsilon \nabla h_0 \cdot \nabla u dX = \int k_0 \zeta_\varepsilon u dX + u(p_i) - \int (u - u(p_i)) \nabla h_0 \cdot \nabla \zeta_\varepsilon dX.$$

The flux of $-\log \rho / (2\pi)$ through the small annulus thus produces $u(p_i)$, and by Step 2 the last term vanishes in the limit. Altogether $\langle h_0, v \rangle = u(p_i) + \mathcal{F}(u)$. On the other hand $\langle w, v \rangle = q_\eta(w, u) = \mathcal{F}(u)$. Hence $\langle g_i, (H_F - \eta)u \rangle = u(p_i)$ for every $u \in \text{Dom } H_F$, which is the defining property of $G_{\eta,i} = (\tau_i R_\eta)^* 1$. Since $H_F - \eta$ is onto, $g_i = G_{\eta,i}$.

Step 4: the expansion. Near p_i , $G_{\eta,i} = -\log \rho / (2\pi) - w$, and near p_j , $j \neq i$, $G_{\eta,i} = -w$. Since w is real and Hölder continuous, this gives the expansion in ρ with $(B_\eta)_{ji} = -w(p_j)$ and $\beta = \alpha$. For symmetry, let $\delta_{j,\varepsilon}$ be the indicator of the ε -disc about p_j , normalized to have integral 1. The sup bound of Theorem 5.1 shows that $R_\eta \delta_{i,\varepsilon}$ is bounded in $\mathcal{H}$, and $\langle R_\eta \delta_{i,\varepsilon}, (H_F - \eta)u \rangle \rightarrow u(p_i)$, so $R_\eta \delta_{i,\varepsilon} \rightarrow G_{\eta,i}$ weakly, and uniformly near p_j by the interior Hölder estimate. Self-adjointness of R_η then gives $G_{\eta,i}(p_j) = G_{\eta,j}(p_i)$. Finally Corollary 3.4 and Lemma 3.2 replace $\log \rho$ by $\log d_g$ or $\log \rho_e$ with errors $O(\rho^{1/2})$, so the same matrix B_η serves with $\beta = \min(\alpha, \frac{1}{2})$. $\square$

Remark 6.2. In the language of conical points [2], a cross-cap has total angle 2π : by Lemma 3.2 the circle of radius r about it has length $2\pi r(1 + o(1))$. This is the borderline case with a single logarithmic channel. Its Whitney invariants do not affect the leading coefficient; they enter only through the regular matrix B_η , together with the global geometry.

Theorem 6.3 (Geometric boundary coordinates). *Every $\varphi \in \text{Dom } A^*$ has unique coefficients $\ell, b \in \mathbb{C}^k$ with*

$$\varphi = \ell_i \log d_g(\cdot, p_i) + b_i + O(d_g^\beta) \quad \text{at } p_i,$$

namely $\ell = -a/(2\pi)$ and $b = \tau u + B_\eta a$ for $\varphi = u + G_\eta a$. The map $\varphi \mapsto (\ell, b)$ is graph-bounded, onto $\mathbb{C}^k \oplus \mathbb{C}^k$, and has kernel $\text{Dom } A$. If ψ has coefficients (ℓ', b') , then

$$\langle A^* \varphi, \psi \rangle - \langle \varphi, A^* \psi \rangle = 2\pi(\langle \ell, b' \rangle - \langle b, \ell' \rangle).$$

The Friedrichs domain is $\{\ell = 0\}$, and the self-adjoint extensions are the maximal isotropic planes of this form, namely $(U - I)b + i(U + I)a = 0$ with $a = -2\pi\ell$ and $U \in U(k)$. For a Hermitian matrix C , the extension $b = Ca$ has resolvent $R_z + G_z(C - B_\eta - M_z)^{-1}G_z^$, where $M_z = (z - \eta)\tau R_z G_\eta$.*

Proof. Substitute Theorem 6.1 into the decomposition of Theorem 5.2. Uniqueness follows by dividing by the logarithm. The pairing changes from $\langle \tau u, a' \rangle - \langle a, \tau u' \rangle$ by the symmetric shift B_η , which cancels. The resolvent formula follows from $G_z - G_\eta = (z - \eta)R_z G_\eta \in \text{Dom } H_F$. $\square$

7 Uniqueness, scale and sheets

Corollary 7.1 (Markov uniqueness). *H_F is the only self-adjoint extension of A whose semigroup is sub-Markovian.*

Proof. q is a Dirichlet form, so H_F is sub-Markovian. If H is another sub-Markovian extension, then $\|(H + \lambda)^{-1}h\|_\infty \leq \lambda^{-1}\|h\|_\infty$ for bounded $h \in \mathcal{H}$. By Theorem 6.3, a nonzero logarithmic coefficient would make $(H + \lambda)^{-1}h$ unbounded near its pole, since the density is bounded below in arclength coordinates. All coefficients therefore vanish, the two resolvents agree on a dense set, and $H = H_F$. $\square$

The same argument shows that, in the plane, the free Laplacian is the only Markovian point interaction with respect to Lebesgue measure. Dirichlet forms associated with point interactions are studied in [11].

Corollary 7.2 (Reference length). *Replacing $\log d_g$ by $\log(d_g/\lambda_i)$ changes b to $b + \text{diag}(\log \lambda_i)\ell$. The only self-adjoint boundary plane invariant under every common change $\lambda_i \mapsto e^t \lambda_i$ is the Friedrichs plane $\ell = 0$.*

Proof. If (ℓ, b) lies in an invariant plane, so does $(0, \ell)$. Pairing the two gives $2\pi\|\ell\|^2 = 0$. $\square$

Corollary 7.3 (Sheet-even coefficients). *If $s_\varepsilon \neq t_\varepsilon$ tend to the same p_i with $f(s_\varepsilon) = f(t_\varepsilon)$, then for every $\varphi \in \text{Dom } A^*$, $\varphi(s_\varepsilon) - \varphi(t_\varepsilon) = O(\rho_\varepsilon^\beta) \rightarrow 0$. The coefficients are the same on both sheets.*

Proof. The two points have the same extrinsic radius, so the expansions in $\rho_{e,i}$ agree up to the remainders. $\square$

8 Why the boundary must be fixed

Let H_0 be the graph closure of $-\Delta_g$ on $C_c^\infty(\text{int } \Sigma \setminus Z)$ in case (D).

Proposition 8.1. (1) *For every $z \in \rho(H_F)$, $\ker(H_0^* - z)$ is infinite-dimensional. In particular the deficiency spaces of H_0 are infinite-dimensional.*

(2) *H_0 has at least two distinct sub-Markovian self-adjoint extensions.*

(3) *$H_0 \geq \lambda_D > 0$, and its Friedrichs extension is H_F . Its Krein-von Neumann extension $H_{0,K}$ has*

$$\text{Dom } H_{0,K} = \text{Dom } H_0 \dot{+} \ker H_0^*, \quad \ker H_{0,K} = \ker H_0^*,$$

which is infinite-dimensional. The part of $H_{0,K}$ in $(\ker H_{0,K})^\perp$ has purely discrete spectrum, and $\lambda \neq 0$ is one of its eigenvalues if and only if the weak buckling problem

$$\langle H_0 v, H_0 \varphi \rangle = \lambda \langle v, H_0 \varphi \rangle \quad (\varphi \in \text{Dom } H_0)$$

has a solution $0 \neq v \in \text{Dom } H_0$.

Proof. (1) Take a regular boundary arc. For smooth boundary data ϕ supported in it, let w_ϕ be a smooth collar extension, supported away from $Z \cup Z_\partial$, and put $v_\phi = w_\phi - (H_F - z)^{-1}(-\Delta_g - z)w_\phi$. Then $(-\Delta_g - z)v_\phi = 0$ in the interior, so $v_\phi \in \ker(H_0^* - z)$, and the trace of v_ϕ on the arc is ϕ . Independent ϕ give independent v_ϕ .

(2) H_F is one sub-Markovian extension. Closing q on smooth functions with free boundary values gives another, the Neumann Laplacian, whose kernel contains the constants.

(3) H_0 is closed, densely defined and contained in H_F , so $H_0 \geq \lambda_D$ by Theorem 4.2. Its form core $C_c^\infty(\text{int } \Sigma \setminus Z)$ is a core for q by Proposition 4.1, so its Friedrichs extension is H_F . The Krein domain and kernel are given by [10, Thm. 2.1], and (1) with $z = 0 \in \rho(H_F)$ shows that the kernel is infinite-dimensional. Since H_F has compact resolvent, the abstract results [10, Hyp. 2.2, Thms. 2.4 and 3.4] give the discreteness of the reduced spectrum and the buckling characterization. $\square$

The finite classification of Section 5 therefore requires the outer boundary condition to be part of the definition of the operator. With it, the point-restricted operator A has a Krein extension whose kernel $G_0 \mathbb{C}^k$ has dimension exactly k (Theorem 5.2).

9 Examples

9.1 Steiner's Roman surface

Let $\Sigma = \mathbb{RP}^2$ and let $f[x : y : z] = (yz, zx, xy)$ on unit representatives. This is case (N).

Proposition 9.1. *f is an immersion except at the six points $[1 : \pm 1 : 0]$, $[1 : 0 : \pm 1]$ and $[0 : 1 : \pm 1]$, each of which is a Whitney cross-cap. In an orthonormal adapted frame, the Whitney determinant (1) equals ± 2 .*

Proof. On S^2 the differential is the restriction of $M = \begin{pmatrix} 0 & z & y \\ z & 0 & x \\ y & x & 0 \end{pmatrix}$, and $\det M = 2xyz$. The restriction to the tangent plane has rank below two only if M has a kernel vector tangent to the sphere. This forces one coordinate to vanish and the other two to be equal up to sign, which gives twelve points on S^2 and six on $\mathbb{RP}^2$. With η the tangent kernel vector, $\xi = p \times \eta$ and the chart $(s, t) \mapsto (p + s\xi + t\eta)/|p + s\xi + t\eta|$, an exact computation (replayed in the verifier) gives $f_t = 0$, $|f_s| = 1$ and $\det(f_s, f_{st}, f_{tt}) = \pm 2$. $\square$

Theorems 5.1–6.3 therefore give a $U(6)$ -family of pseudo-Laplacians on the Roman surface, each with six logarithmic channels. The image has three double segments along the coordinate axes, and the six cross-caps are their endpoints $\pm \frac{1}{2}e_k$ (Figure 3).

Theorem 9.2 (Symmetry of the Green matrix). *Index the cross-caps so that $P_\perp$ pairs each with the unique one whose line is orthogonal to it, and let $P_\circ = J - I - P_\perp$ be the adjacency of the remaining pairs, whose lines meet at 60° . For every $\eta < 0$ there are real numbers $r, \gamma_\perp, \gamma_\circ$ with*

$$B_\eta = rI + \gamma_\perp P_\perp + \gamma_\circ P_\circ.$$

Hence B_η has at most three eigenvalues: $r + \gamma_\perp + 4\gamma_\circ$ on the constants, $r + \gamma_\perp - 2\gamma_\circ$ with multiplicity two, and $r - \gamma_\perp$ with multiplicity three.

(a) Roman surface: double segments, six cross-caps

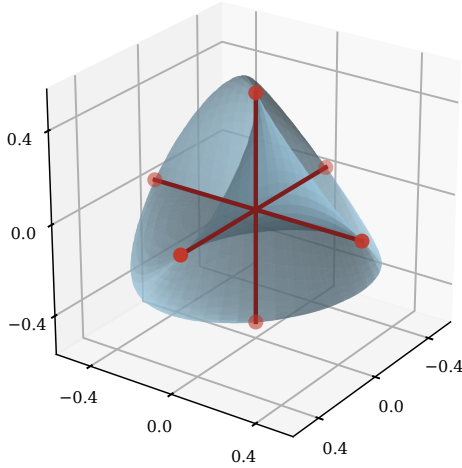
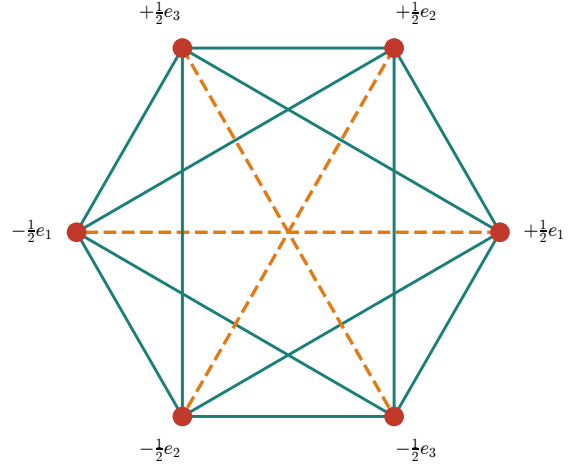
(b) P_o (solid, octahedron) and $P_\perp$ (dashed)

Figure 3: (a) Steiner's Roman surface $f[x : y : z] = (yz, zx, xy)$. Its three double segments (dark red) end at the six cross-caps $\pm\frac{1}{2}e_k$. (b) The pair structure of Theorem 9.2. The two ends of one double segment form an orthogonal pair (dashed, the matrix $P_\perp$); all other pairs meet at 60° in the source and form the edges of an octahedron (solid, P_o).

Proof. Each signed permutation σ of coordinates descends to an isometry of $(\mathbb{RP}^2, g)$, because $f \circ \sigma = L_\sigma \circ f$ for a signed permutation L_σ of $\mathbb{R}^3$, which is a Euclidean isometry. The pullback by σ is unitary, commutes with H_F and R_η , permutes the cross-caps, and preserves extrinsic radii. Hence it permutes the Green vectors, and $(B_\eta)_{\sigma i \sigma j} = (B_\eta)_{ij}$. This group acts transitively on the six cross-caps, and its orbits on ordered pairs of distinct cross-caps are exactly the six orthogonal pairs and the twenty-four 60° pairs (verified). Symmetry of B_η completes the form. $P_\perp$ and P_o commute; P_o is the adjacency matrix of the octahedron. Their joint eigenvalues are $(1, 4)$ on the constants, $(1, -2)$ on matching-even vectors of sum zero, and $(-1, 0)$ on matching-odd vectors. $\square$

9.2 A ruled surface with boundary cross-caps

The ruled map

$$S(t, u) = (\cos t + 2u(1 - \cos t), (1 - u) \sin t, u\sqrt{\cos t(2 - \cos t)})$$

on $[-\pi/2, \pi/2] \times [0, 1]$, completed at $t = \pm\pi/2$ by the chart $r = Q$, arose in a two-component signal model [13, 12]. It has five rank-one points, all Whitney cross-caps [12]. Two are interior: $P_\pm = (\pm t_0, u_0)$ with $\cos t_0 = (3 - \sqrt{5})/2$ and $u_0 = (1 - \cos t_0)/2$. Three lie on the boundary: a half-germ at $(0, 1)$ and two quarter-germs at the lower corners. In the completed charts the boundary is formed by coordinate lines, so Definition 2.1 holds in case (D) with $k = 2$. All results apply. Moreover $S(-t, u) = JS(t, u)$ with $J = \text{diag}(1, -1, 1)$ (Figure 4), and the argument of Theorem 9.2 shows $B_\eta = \begin{pmatrix} r & \gamma \\ \gamma & r \end{pmatrix}$. The Krein extension has a two-dimensional kernel. An earlier preprint on this surface [14] defined the two-point problem through the compact-core closure H_0 and claimed indices $(2, 2)$; Proposition 8.1 shows why the outer boundary must be fixed first.

10 Open problems

- *Explicit regular parts.* Compute or estimate B_η , for instance the three Roman-surface parameters, and relate them to global geometry.

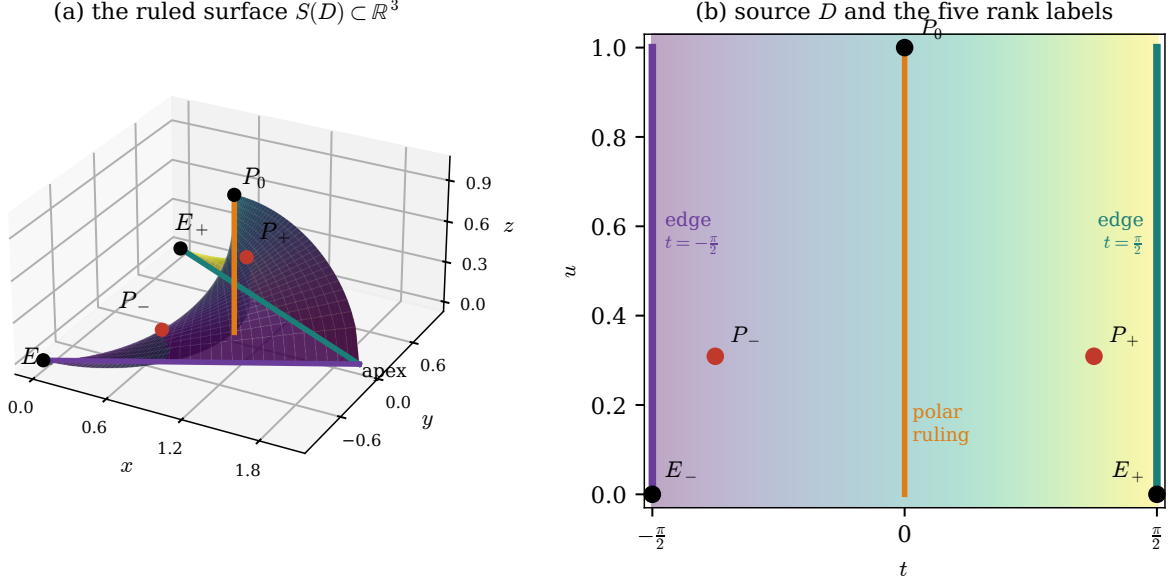


Figure 4: The ruled surface of Section 9.2 and its source. The interior cross-caps $P_{\pm}$ carry the two logarithmic channels; P_0 is a boundary half-germ and $E_{\pm}$ are corner quarter-germs, all on the Dirichlet boundary.

- *Sharp local geometry.* For the standard germ $G - I = O(\rho \log(1/\rho))$ (Remark 3.3). Is this the rate for every Whitney germ, and do the Whitney invariants appear in the next term?
- *Spectral asymptotics.* By Corollary 5.3, a Weyl law for H_F would hold for every extension. Establishing it for the degenerate metric, and finding the heat-trace contribution of each cross-cap, remain open.
- *Dirac operators.* A spinor or twisted Dirac analogue at cross-caps needs closed domains, cut traces and transmission conditions. The square-root monodromy of the observation field of the ruled surface suggests a natural twist [13].
- *Other singularities.* Non-generic germs such as cuspidal edges or $S_k^{\pm}$ singularities may carry different channel structures.

A Evidence

The proofs above are given in full. The figures are generated by a script from the explicit formulas of the paper; Figure 2 is a numerical illustration of Remark 3.3. The external inputs are the cross-cap recognition criterion [6], Whitney’s genericity theorem [4, 5], the scalar Hölder estimates [8, 9], and the abstract extension and Krein theorems [7, 10]. An exact replay checks the algebra the proofs rely on: the adapted metric and determinant identities; the explicit arclength coordinate of Remark 3.3 and its derivative identities; the singular set and Whitney determinants of the Roman surface; its symmetry orbits on pairs of cross-caps; the eigenvalue pattern of $rI + \gamma_{\perp}P_{\perp} + \gamma_{\circ}P_{\circ}$; and the reflection identity of the ruled surface. These checks do not certify the analysis, and no Lean formalization of the analytic statements is claimed.

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